

Francis Mars

Fullstack Software Engineer — Real-Time Systems, Bitcoin & Lightning Network

francismars.com · github.com/francismars · x.com/btcfrancis

PROFILE

Fullstack engineer specializing in real-time systems and Bitcoin payment infrastructure. Built and shipped production apps on Lightning Network and Nostr — from multiplayer games to conference-grade live payment displays. Regular speaker at major Bitcoin events. Stack: React/Next.js, Node.js/TypeScript, Rust, Go, React Native.

PROJECTS

MarsPay — Real-Time Payment Backend · github.com/francismars/marspay

Node.js · TypeScript · Express · Socket.IO · Lightning (LNbits, LNURL) · Nostr (NDK)

- Authoritative server architecture for real-time paid apps (games, duels, tournaments) — server owns state, clients send inputs
- Lightning integration for deposits, withdrawals, webhook settlement; Nostr (NDK) for zaps; defines Socket.IO protocol used by multiple clients

PubPay — Decentralized Social Payments · pubpay.me

React · Nostr (NDK, NWC) · Lightning · Monorepo

- **PubPay** — shareable payment requests as social posts (no sign-ups, no fees); **PubPay Live** — real-time zap display for live events; **PubPay Multi Live** — multi-stage conference tool with timeline editor, auto-rotation, and rehearsal mode
- Battle-tested at Bitcoin 2025, BCC8333, Adopting Bitcoin

Chain Duel — Real-Time Multiplayer Bitcoin Game · chainduel.net

React · PixiJS · Zustand · WebSockets · Lightning · Nostr

- Custom game engine: PixiJS rendering, Zustand state, keyboard/gamepad input, audio pipeline; payment-gated matches via MarsPay
- Live Lightning Network demo at Bitcoin conferences worldwide — 3 years of production conference use

Homeserver Dashboard — Next.js Admin UI · github.com/francismars/homeserver-dashboard

Next.js · Tailwind · shadcn/ui · Docker · Tailscale (Pubky engagement)

- User management, WebDAV file browser, API explorer; Docker/Umbrel integration; Tailscale tunnel for closed networks

TECH STACK

Frontend	React, Next.js, PixiJS, Zustand, Tailwind, shadcn/ui, React Native
Backend	Node.js, TypeScript, Express, Socket.IO, WebSockets
Payments	Lightning (LNbits, LNURL, NWC), Nostr (NDK), BOLT 11/12
Systems	Rust, Go, Docker, Linux

Open Source: Pubky · Satsigner · Stable Channels · Inurlp (LNbits) · Nigiri Bitcoin · go-elements · tdex-feeder

Speaking: Bitcoin Las Vegas · Bitcoin Amsterdam · Plan B Forum · Adopting Bitcoin · Bitcoin Atlantis · BTC Balkans · Satsconf · *and others*

Education: M.Sc. Informatics, University of Lisbon (17/20) · B.Sc. Information Technologies, University of Lisbon

Training: Chaincode Labs — Bitcoin Protocol (2021) · Lightning Protocol (2022) · Sovereign Engineering Cohort, Madeira (2024)